

MACHINE LEARNING-BASED EARTHQUAKE MAGNITUDE PREDICTION

Research Paper

Team-6

Rahul (2420090043) | Sai Sujith (2420030280) | Nikhileash (2420030596) | Sumanth Reddy (2420030782)

Department of Computer Science and Engineering
Koneru Lakshmaiah Education Foundation

Abstract

Earthquakes are destructive natural hazards that can cause severe damage to people, infrastructure, and essential services. This research presents a machine learning-based framework for earthquake magnitude prediction using historical seismic information. The proposed system considers commonly available parameters including latitude, longitude, depth, time, and seismic measurements. The study investigates and compares Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost, and Artificial Neural Network (ANN) models. The data-processing pipeline includes data collection, cleaning, missing-value handling, feature selection, normalization, model training, prediction, and evaluation. Model performance is intended to be assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. A practical application layer is also proposed in which predicted magnitude is converted into risk categories and warnings, with an SMS-based notification mechanism for high-risk events. The research focuses on building a reproducible and deployable framework rather than claiming a measured model winner before the final experimental runs are completed.

Keywords—Earthquake Magnitude Prediction, Machine Learning, Seismic Data, Random Forest, XGBoost, Artificial Neural Network, Risk Classification, SMS Alert

1. Introduction

Earthquakes are sudden releases of energy within the Earth's crust that generate seismic waves. Their impact depends on factors such as magnitude, depth, location, distance from populated regions, and local conditions. Because earthquake events can occur without convenient warning, improving the analysis of historical seismic patterns is an important research problem in disaster preparedness.

Conventional statistical approaches can have difficulty representing complex and nonlinear relationships among seismic variables. Machine learning provides a data-driven alternative by learning relationships between historical earthquake parameters and observed magnitude values. Different algorithms offer different trade-offs between interpretability, computational requirements, and their ability to model nonlinear relationships.

This work proposes an Earthquake Magnitude Prediction system that accepts historical or observed seismic parameters and estimates earthquake magnitude. The project also extends the prediction task into a practical risk-warning workflow. The proposed application can display the predicted magnitude and corresponding risk category and can be connected to an SMS service for notification. The project presentation identifies USGS Earthquake Catalog, Kaggle earthquake data, IRIS Seismic Data, NOAA earthquake data, and Global Earthquake Model (GEM) as candidate data sources.

2. Problem Statement

Traditional statistical methods may struggle with complex and nonlinear seismic patterns. There is therefore a need for a machine learning framework that can learn from historical seismic data, compare multiple prediction algorithms, evaluate their errors using several metrics, and present the output in a form that can support preparedness and disaster-management activities.

3. Objectives

- Predict earthquake magnitude from historical seismic parameters.
- Analyze and preprocess historical earthquake data.
- Compare Linear Regression, Decision Tree, Random Forest, XGBoost, and ANN models.

- Evaluate the models using MAE, MSE, RMSE, and R^2 Score.
- Identify a model that provides an appropriate balance between prediction performance and computational cost after experimental evaluation.
- Develop a user-friendly Earthquake Predictor that accepts seismic parameters and displays a predicted magnitude.
- Convert predicted magnitude into risk categories and warnings.
- Provide a practical SMS/mobile alert mechanism for high-risk predictions.

4. Related Work / Literature Review

Previous research has explored classical machine learning, ensemble learning, recurrent neural networks, convolutional networks, and attention-based deep learning for earthquake-related prediction tasks. The reviewed studies indicate that performance can depend strongly on geographic region, dataset size, selected seismic indicators, waveform availability, and the definition of the prediction task.

No.	Research Paper	Model / Algorithm	Dataset	Key Outcome	Limitation
1	Recent Advances in Early Earthquake Magnitude Estimation by Using Machine Learning Algorithms: A Systematic Review	SVM, CNN, CNN-LSTM, XGBoost, RF, LSTM	K-NET, KiK-net, STEAD, IPOC and others	Reviews multiple ML/DL approaches for earthquake magnitude estimation.	Performance varies with dataset, region, and waveform quality.
2	A machine learning framework with parameter optimization and regional clustering for strong earthquake prediction	Random Forest	Japan and New Zealand earthquake catalogs	Uses regional clustering and multiple seismic activity indicators.	Requires engineered seismic indicators and regional clustering; generalization may be limited.
3	Climatic and seismic data-driven deep learning model for earthquake magnitude prediction	Deep Neural Network / Deep Learning	Indonesia, Japan and HKKH-region datasets	Combines climatic and seismic information for magnitude prediction.	Additional climatic variables increase data requirements and model complexity.
4	Scalable earthquake magnitude prediction using spatio-temporal data and model versioning	LightGBM, XGBoost, Gradient Boosting, Random Forest, Linear Regression	Spatio-temporal earthquake dataset	Compares scalable ML models with an MLOps-oriented workflow.	Performance depends on dataset size and feature quality.
5	Earthquake Prediction Using Machine Learning: Improved Magnitude Estimation with LSTM over Random Forest	Random Forest + LSTM	Historical latitude, longitude, depth and magnitude	Uses LSTM to address sequential/time-series aspects of earthquake data.	LSTM requires sequential data preparation and greater computational complexity.
6	Comparative Analysis of Decision Tree, Random Forest, SVM, and Neural Network Models for Predicting Earthquake Magnitude	Decision Tree, Random Forest, SVM, Neural Network	USGS earthquake dataset	Compares several ML models using MAE, RMSE and R^2 .	Results depend on the selected USGS data and geographic coverage.
7	Application of Machine Learning Models to Multi-Parameter Maximum Magnitude Prediction	Random Forest, LSTM and other ML models	Southern China North–South Seismic Belt	Uses multiple seismicity indicators and compares several ML approaches.	Regional focus may limit transferability.
8	Machine Learning Applications for Earthquake Magnitude Prediction in Western Turkey	LSTM, CNN, ANFIS, Random Forest, Decision Tree	Western Turkey earthquake dataset	Compares ML/DL methods using several error metrics.	Focuses on a specific region and weekly average magnitude.
9	Deep learning model for predicting earthquake magnitude using bidirectional	Bi-LSTM + Attention	2,000 records from Mediterranean, Southern Europe, North Africa and	Reports high performance for its defined task and	Relatively limited dataset and focus on earthquakes above

	LSTM with attention mechanism on historical seismic data from 2017 to 2025		Western Asia	dataset.	magnitude 4.9.
10	Forecasting earthquakes by Machine Learning techniques: lights and shadows	Logistic Regression, Random Forest, XGBoost, LSTM	Southern California Earthquake Data Center catalog	Compares multiple ML approaches for earthquake forecasting.	Results are based on Southern California data and require validation elsewhere.

5. Research Gap and Motivation

No.	Research Gap	How the Proposed Project Addresses It
1	Region-specific models	Use a larger and geographically diverse earthquake dataset to improve generalization.
2	Dependence on complex waveform data	Use commonly available parameters such as latitude, longitude, depth, time, and seismic measurements.
3	High computational cost of deep learning	Compare traditional and advanced models and consider both predictive performance and computational cost.
4	Limited algorithm comparison	Compare Linear Regression, Decision Tree, Random Forest, XGBoost, and ANN.
5	Data-quality effects	Apply data cleaning, missing-value handling, feature selection, and normalization.
6	Limited disaster-management support	Add a risk-level and alert mechanism based on predicted magnitude.
7	Evaluation using limited metrics	Use MAE, MSE, RMSE, and R ² Score for a broader evaluation.
8	Complex models may be difficult to interpret	Include interpretable models such as Decision Tree and Random Forest in the comparison.
9	Limited real-time application focus	Develop a user-facing Earthquake Predictor for seismic parameter input and magnitude output.
10	Prediction without actionable information	Translate predicted magnitude into risk categories and warnings.

6. Dataset and Data Sources

The project identifies several possible sources for earthquake data: the USGS Earthquake Catalog, Kaggle Earthquake Database, IRIS Seismic Data, NOAA Earthquake Database, and the Global Earthquake Model (GEM). The final implementation should document the exact dataset selected, collection period, geographic coverage, number of records, target variable, and preprocessing decisions so that the reported experiments can be reproduced.

Candidate input features include:

- Latitude — geographic latitude of the earthquake occurrence.
- Longitude — geographic longitude of the earthquake occurrence.
- Depth — depth below Earth's surface at which the earthquake starts.
- Time — date and time associated with the seismic event.
- Seismic measurements — recorded earthquake-related measurements available in the selected dataset.

The target variable is earthquake magnitude. Before model training, the selected dataset should be checked for missing values, duplicated records, inconsistent units, invalid measurements, and unsuitable fields. Feature selection and normalization should be applied according to the requirements of the selected algorithms.

7. Proposed Methodology

The proposed methodology follows a sequence from data collection to deployment. The overall pipeline is: Dataset Collection → Data Cleaning → Missing-Value Handling → Feature Selection → Normalization → Train/Test Split → Model Training → Magnitude Prediction → Evaluation → Risk Classification → Alert/Dashboard.

7.1 Data Preprocessing

- Collect historical earthquake records from the selected data source.
- Remove duplicate or invalid records.
- Handle missing values using a documented strategy appropriate to each feature.
- Select relevant earthquake parameters and define magnitude as the prediction target.
- Normalize or standardize numerical features when required.
- Split the processed dataset into training and testing subsets using a documented random seed and split ratio.

7.2 Machine Learning Models

No.	Model	Role in the Project
1	Linear Regression	Provides a baseline by learning a linear relationship between seismic input features and magnitude.
2	Decision Tree Regressor	Uses hierarchical decision rules to model nonlinear relationships and is relatively interpretable.
3	Random Forest Regressor	Combines multiple decision trees to reduce dependence on a single tree and model nonlinear patterns.
4	XGBoost	Uses gradient boosting to build an ensemble of decision trees and can model complex feature relationships.
5	Artificial Neural Network (ANN)	Learns nonlinear relationships through interconnected layers and can represent more complex patterns.

7.3 Model Evaluation

No.	Metric	Description
1	Mean Absolute Error (MAE)	Average absolute difference between actual and predicted magnitudes. Lower values indicate smaller average errors.
2	Mean Squared Error (MSE)	Average squared prediction error. Larger errors receive greater penalty.
3	Root Mean Squared Error (RMSE)	Square root of MSE; expresses error in the same unit as the magnitude target.
4	R ² Score	Measures the proportion of variation in the target explained by the model; values closer to 1 generally indicate better explanatory performance.

8. System Architecture and Application Design

The proposed system is organized into separate modules so that data processing, model training, prediction, risk classification, and user interaction can be developed and evaluated independently.

- Data Collection Module — acquires historical earthquake records and prepares them for processing.
- Preprocessing Module — performs cleaning, missing-value handling, feature selection, and normalization.
- Model Training Module — trains the selected regression models.
- Prediction Module — accepts seismic parameters and generates predicted magnitude.

- Evaluation Module — calculates MAE, MSE, RMSE, and R² Score.
- Risk Classification Module — maps predicted magnitude to predefined project risk categories.
- Alert Module — sends an SMS/mobile notification when a configured high-risk condition is reached.
- Monitoring Interface — Streamlit can be used to provide a simple dashboard for input, prediction, and results.

9. Risk-Level and SMS Alert Mechanism

The project proposes an additional practical layer that converts a predicted magnitude into a risk category. The exact category boundaries should be defined and documented by the project team before deployment rather than being assumed to be universal, because earthquake impact is not determined by magnitude alone. The application can display the predicted magnitude, risk category, and a preparedness message. For a configured high-risk condition, the alert module can send an SMS through the Twilio SMS API. This mechanism is intended as a decision-support and notification feature, not as a claim of guaranteed real-time earthquake prediction.

10. Tools and Technologies

Technology	Role
Python	Core development and machine learning implementation.
Scikit-learn / TensorFlow	Machine learning model development and evaluation.
Pandas and NumPy	Data preprocessing, transformation, and numerical analysis.
IoT Seismic Sensors	Potential source of real-time seismic measurements.
GPS/GIS	Location tracking and affected-area mapping.
Twilio SMS API	Potential SMS warning delivery mechanism.
MySQL / Firebase	Potential data storage and user management.
Streamlit	Interactive monitoring and prediction dashboard.
Google Colab	Model development and experimentation.
AWS / GCP Cloud	Potential cloud deployment and scalable processing.

11. Experimental Plan

The final experimental study should train all selected models on the same cleaned dataset and use the same training/testing protocol wherever applicable. This controlled procedure is necessary for a fair comparison. The experiment should record training time, prediction time where relevant, model configuration, feature set, dataset size, and evaluation results.

- Finalize and document the earthquake dataset and its geographic/time coverage.
- Perform exploratory data analysis to identify distributions, missing values, outliers, and feature relationships.
- Prepare the training and testing datasets.
- Train Linear Regression, Decision Tree, Random Forest, XGBoost, and ANN models.
- Calculate MAE, MSE, RMSE, and R² Score for each model.
- Generate comparison tables and plots for the evaluation metrics.
- Analyze errors and investigate whether geographic or temporal subsets behave differently.
- Select a model for the application only after reviewing both performance and computational requirements.
- Integrate the selected model with the prediction dashboard and risk/alert mechanism.

12. Results and Discussion

At the time of preparation of this paper, the supplied project material defines the proposed models, features, evaluation metrics, and application workflow but does not provide a completed experimental results table for the earthquake dataset. Therefore, numerical model-performance values are intentionally not fabricated here.

The final results section should report, for each model, MAE, MSE, RMSE, and R² Score on the same held-out test set. A model-comparison table should be inserted after the final experiments. The discussion should identify which models capture nonlinear relationships effectively, whether the ensemble methods improve error metrics, how ANN training cost

compares with classical models, and whether the selected model is suitable for the intended application. Any claim about the 'best' model should be based directly on the measured experimental results.

Model	MAE	MSE	RMSE	R ² Score	Training/Inference Notes
Linear Regression	To be measured	To be measured	To be measured	To be measured	To be measured
Decision Tree	To be measured	To be measured	To be measured	To be measured	To be measured
Random Forest	To be measured	To be measured	To be measured	To be measured	To be measured
XGBoost	To be measured	To be measured	To be measured	To be measured	To be measured
ANN	To be measured	To be measured	To be measured	To be measured	To be measured

13. Expected Outcomes

- A trained machine learning system capable of estimating earthquake magnitude from selected seismic parameters.
- A quantitative comparison of five machine learning approaches.
- Identification of a suitable model after evaluation using MAE, MSE, RMSE, and R² Score.
- A user-friendly Earthquake Predictor interface.
- A risk-level classification and warning workflow.
- A potential SMS-based notification mechanism for configured high-risk predictions.
- A framework that can be extended with real-time sensor feeds and cloud deployment.

14. Limitations

- Magnitude prediction performance depends on the quality, coverage, and consistency of the selected earthquake dataset.
- A model trained on a particular geographic region may not generalize equally to different tectonic regions.
- Magnitude alone does not fully determine earthquake damage or human impact.
- Real-time sensor integration introduces additional latency, calibration, noise, and infrastructure requirements.
- The SMS mechanism depends on external communication services and cannot guarantee delivery.
- The supplied project material does not yet contain completed experimental results, so model performance cannot be stated as a measured fact in this draft.

15. Future Work

- Use larger and geographically diverse datasets to improve generalization.
- Integrate real-time seismic sensor data where reliable sensor infrastructure is available.
- Investigate time-series models such as LSTM and Bi-LSTM for sequential seismic information.
- Explore CNN or attention-based models when waveform or richer temporal data are available.
- Apply explainability methods to identify which seismic features contribute most to predictions.
- Use cross-validation or repeated experiments to estimate performance variability.
- Deploy the prediction service on AWS/GCP for scalable processing.
- Enhance the dashboard with GIS-based affected-area visualization.
- Calibrate risk thresholds and warning messages using domain-appropriate evidence and validation.

16. Conclusion

This paper proposes a machine learning-based framework for earthquake magnitude prediction using historical seismic data. The project focuses on commonly available parameters such as latitude, longitude, depth, time, and seismic measurements and compares Linear Regression, Decision Tree, Random Forest, XGBoost, and ANN models. The methodology emphasizes systematic preprocessing and evaluation through MAE, MSE, RMSE, and R² Score. Beyond numerical prediction, the proposed application converts model output into risk information and provides a pathway for SMS-based warnings and dashboard-based monitoring. The final model selection and performance claims must be based on the completed controlled experiments. This staged approach provides a practical foundation for developing an earthquake-analysis system while clearly separating proposed functionality from experimentally verified results.

References

- [1] “Recent Advances in Early Earthquake Magnitude Estimation by Using Machine Learning Algorithms: A Systematic Review,” MDPI.
- [2] “A machine learning framework with parameter optimization and regional clustering for strong earthquake prediction,” Springer.
- [3] “Climatic and seismic data-driven deep learning model for earthquake magnitude prediction,” Frontiers.
- [4] “Scalable earthquake magnitude prediction using spatio-temporal data and model versioning,” Nature / PubMed Central.
- [5] “Earthquake Prediction Using Machine Learning: Improved Magnitude Estimation with LSTM over Random Forest,” Zenodo.
- [6] “Comparative Analysis of Decision Tree, Random Forest, SVM, and Neural Network Models for Predicting Earthquake Magnitude,” USGS-based comparative study.
- [7] “Application of Machine Learning Models to Multi-Parameter Maximum Magnitude Prediction,” study of the North–South Seismic Belt.
- [8] “Machine Learning Applications for Earthquake Magnitude Prediction in Western Turkey,” Western Turkey earthquake dataset study.
- [9] “Deep learning model for predicting earthquake magnitude using bidirectional LSTM with attention mechanism on historical seismic data from 2017 to 2025,” historical regional earthquake study.
- [10] “Forecasting earthquakes by Machine Learning techniques: lights and shadows,” Southern California Earthquake Data Center catalog study.

Appendix A — Suggested Result Table for Final Experiment

Before final submission, replace the placeholders in Section 12 with the actual output of the executed experiments. Include the dataset name, number of records, train/test split, preprocessing settings, random seed, model hyperparameters, and complete metric values.